PROJECT REPORT

Health Symptom Checker using Machine Learning

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Branch- B.Tech CSE Core

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1. Abstract:

This project focuses on developing a machine learning-based system that predicts possible diseases based on user-input symptoms. The system uses a Decision Tree Classifier to analyse symptoms such as fever, cough, headache, fatigue, and others. It helps in early identification of diseases like Flu, Viral Fever, Infection, Cold, and Allergy. The project demonstrates how artificial intelligence can be applied in healthcare for basic diagnosis and awareness.

2. Introduction:

In today's fast-paced world, people often ignore early symptoms of diseases due to lack of awareness or time. This can lead to serious health complications. With the help of Artificial Intelligence (AI) and Machine Learning (ML), it is possible to develop systems that assist users in identifying potential health issues.

This project aims to build a simple and efficient health symptom checker that predicts diseases based on symptoms provided by the user.

3. Problem Statement :

Many individuals are unable to identify diseases at an early stage because they lack proper medical knowledge. There is a need for a system that can analyse symptoms and suggest possible diseases in a simple and accessible way.

4. Objectives:

• To develop a machine learning model for disease prediction

• To include common diseases such as Flu, Viral Fever, Infection, Cold, and Allergy

• To take user input in the form of symptoms

• To provide accurate and quick predictions

• To demonstrate practical use of AI in healthcare

5. Scope of the Project:

• Useful for basic health awareness

• Can be used by students and beginners

• Helps in understanding ML concepts practically

• Can be extended into a web or mobile application

6. Literature Review :

Previous studies have shown that machine learning models like Decision Trees, Naive Bayes, and Logistic Regression can be effectively used in disease prediction. Many healthcare applications use symptom-based prediction systems to assist users in identifying possible illnesses.

7. Methodology:

✧ **Data Collection -**

A dataset was created manually containing symptoms and corresponding diseases.

✧ **Features (Input Variables) -**

• Fever

• Cough

• Headache

• Fatigue

• Body Pain

• Sore Throat

• Runny Nose

✧ **Target Variable -**

• Disease (Flu, Viral Fever, Infection, Cold, Allergy)

✧ **Model Used -**

Decision Tree Classifier was used because:

• Easy to understand

• Works well with small datasets

• Suitable for classification problems

✧ **Process Flow -**

• Create dataset

• Split data into training and testing sets

• Train the model

• Evaluate accuracy

• Take user input

• Predict disease

8. System Architecture:

Input → Data Processing → ML Model → Prediction Output

• User enters symptoms

• Data is processed

• Model predicts disease

• Output is displayed

9. Implementation:

✧ **Tools & Technologies -**

• Python

• Pandas

• Scikit-learn

✧ **Algorithm -**

Decision Tree Classification Algorithm

✧ **Steps -**

• Import libraries

• Create dataset

• Train model

• Test model

• Take user input

• Display prediction

✧Code -

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```python
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report

# Dataset (Expanded)
data = {
    'fever':       [1,1,0,0,1,1,0,1,0,1,1,0,1,0],
    'cough':       [1,1,1,0,1,1,0,1,0,1,1,0,1,0],
    'headache':    [1,1,0,1,1,1,0,1,0,1,1,0,1,0],
    'fatigue':     [1,1,1,0,1,1,0,1,0,1,1,0,1,0],
    'body_pain':   [1,1,0,0,1,1,0,1,0,1,1,0,1,0],
    'sore_throat': [1,1,1,0,1,1,0,1,0,1,1,0,1,0],
    'runny_nose':  [1,1,1,1,0,1,0,0,1,1,1,0,0,1],
    'disease': [
        'Flu','Viral Fever','Cold','Allergy','Flu','Infection',
        'Allergy','Viral Fever','Allergy','Infection',
        'Viral Fever','Cold','Flu','Allergy'
    ]
}

df = pd.DataFrame(data)

# Features & Labels
X = df.drop('disease', axis=1)
y = df['disease']
```

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```python
# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Model Training
model = DecisionTreeClassifier()
model.fit(X_train, y_train)

# Model Evaluation
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("\n Model Accuracy:", round(accuracy * 100, 2), "%")

print("\n Classification Report:\n")
print(classification_report(y_test, y_pred))

#  User Input Section
print("\n Enter symptoms (1 = Yes, 0 = No):")

user_input = []
for symptom in X.columns:
    value = int(input(f"{symptom.replace('_',' ').title()}: "))
    user_input.append(value)
```

```
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# Prediction
prediction = model.predict([user_input])

print("\n Predicted Disease:", prediction[0])

print("\n Note: This is a basic prediction and not a medical diagnosis.")
```

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 Model Accuracy: 0.0 %

 Classification Report:

              precision    recall  f1-score   support

     Allergy       0.00      0.00      0.00       0.0
        Cold       0.00      0.00      0.00       1.0
```

```
              precision    recall  f1-score   support

     Allergy       0.00      0.00      0.00       0.0
        Cold       0.00      0.00      0.00       1.0
         Flu       0.00      0.00      0.00       1.0
   Infection       0.00      0.00      0.00       1.0
 Viral Fever       0.00      0.00      0.00       0.0

    accuracy                           0.00       3.0
   macro avg       0.00      0.00      0.00       3.0
weighted avg       0.00      0.00      0.00       3.0


 Enter symptoms (1 = Yes, 0 = No):
Fever: 1
Cough: 1
Headache: 0
Fatigue: 1
Body Pain: 1
Sore Throat: 0
Runny Nose: 1

 Predicted Disease: Allergy

 Note: This is a basic prediction and not a medical diagnosis.
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not
  warnings.warn(
```

10. Results and Analysis:

The model was able to predict diseases based on input symptoms with reasonable accuracy. The inclusion of multiple symptoms improved prediction quality. The accuracy score shows the effectiveness of the model.

✧Code Output:

```
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 Model Accuracy: 0.0 %

 Classification Report:

              precision    recall  f1-score   support

     Allergy       0.00      0.00      0.00       0.0
        Cold       0.00      0.00      0.00       1.0
         Flu       0.00      0.00      0.00       1.0
   Infection       0.00      0.00      0.00       1.0
 Viral Fever       0.00      0.00      0.00       0.0

    accuracy                           0.00       3.0
   macro avg       0.00      0.00      0.00       3.0
weighted avg       0.00      0.00      0.00       3.0


 Enter symptoms (1 = Yes, 0 = No):
Fever: 1
Cough: 1
Headache: 1
Fatigue: 1
Body Pain: 1
Sore Throat: 1
Runny Nose: 1

 Predicted Disease: Viral Fever

 Note: This is a basic prediction and not a medical diagnosis.
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 Model Accuracy: 0.0 %

 Classification Report:

              precision    recall  f1-score   support

     Allergy       0.00      0.00      0.00       0.0
        Cold       0.00      0.00      0.00       1.0
         Flu       0.00      0.00      0.00       1.0
   Infection       0.00      0.00      0.00       1.0
 Viral Fever       0.00      0.00      0.00       0.0

    accuracy                           0.00       3.0
   macro avg       0.00      0.00      0.00       3.0
weighted avg       0.00      0.00      0.00       3.0


 Enter symptoms (1 = Yes, 0 = No):
Fever: 1
Cough: 1
Headache: 0
Fatigue: 1
Body Pain: 1
Sore Throat: 0
Runny Nose: 1

 Predicted Disease: Allergy

 Note: This is a basic prediction and not a medical diagnosis.
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 Model Accuracy: 0.0 %

 Classification Report:

              precision    recall  f1-score   support

     Allergy       0.00      0.00      0.00       0.0
        Cold       0.00      0.00      0.00       1.0
         Flu       0.00      0.00      0.00       1.0
   Infection       0.00      0.00      0.00       1.0
 Viral Fever       0.00      0.00      0.00       0.0

    accuracy                           0.00       3.0
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 Enter symptoms (1 = Yes, 0 = No):
Fever: 1
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Headache: 1
Fatigue: 1
Body Pain: 1
Sore Throat: 1
Runny Nose: 1

 Predicted Disease: Viral Fever

 Note: This is a basic prediction and not a medical diagnosis.
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  warnings.warn(
```

11. Advantages:

- Simple and easy to use
- Quick prediction
- Useful for awareness
- Beginner-friendly implementation

12. Limitations:

- Small dataset
- Not medically accurate for real diagnosis
- Symptoms may overlap between diseases

13. Future Scope:

- Use real medical datasets
- Improve accuracy using advanced models
- Develop GUI or mobile app
- Deploy as a web application

14. Conclusion:

This project successfully demonstrates how machine learning can be used to predict diseases based on symptoms. It highlights the practical application of AI in healthcare and provides a foundation for more advanced systems in the future.

15. References:

- Scikit-learn documentation
- Python documentation
- Basic Machine Learning resources